DESIGN OF A GROUND SOURCE HEAT PUMP FOR COOLING IN A DETACHED RESIDENTIAL BUILDING

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1 INTRODUCTION

1.1 Thermal comfort

Thermal comfort is arguably a fundamental requirement for overall comfort in our homes. It is defined as a mental state in which we are comfortable with the thermal conditions of the environment around us (Wikipedia contributors, Thermal comfort., 2023). There are several factors that can contribute to this mental state like the amount of clothing one has on, the ambient air temperature, the relative humidity of the air and an individual's metabolic rate (Wikipedia contributors, Thermal comfort., 2023). As such modifying the ambient conditions for thermal comfort can be as simple of taking off or adding on articles of clothing. Other approaches include consuming hot or cold food or drinks depending on the conditions, opening, or closing windows and doors depending on the conditions and using fans for forced air circulation (Christopher Tweed, 2014). The solution is not always this simple though and, in many cases, involves using sophisticated heating and cooling systems to add or remove heat from a space. Heat pumps are especially popular for this application.

1.2 Heat pumps

Heat pumps are devices that move heat from on location to another through a refrigeration cycle (Wikipedia contributors, Heat pump., 2023). An air source heat pump in its simplest form consists of a compressor, an evaporator, a condenser, and an expansion valve. The components are all connected using high pressure pipes through which a low boiling point refrigerant flows. In the evaporator, liquid refrigerant colder than the air picks up heat from the air flowing through the evaporator's fins. The refrigerant boils and evaporates then flows out of the evaporator and into the compressor where it is compressed to a higher temperature and pressure. The next stage is the condenser where the now gaseous refrigerant cools and condenses as it loses heat to the colder air flowing over the fins of the condenser. As a high temperature, high pressure liquid, the refrigerant then flows through an expansion valve that lowers its temperature and pressure for the cycle to be repeated. The same system can be used for both heating and cooling by including a reversing valve that reverses the direction of flow of the refrigerant (The Engineering Mindset, 2018).

Ground source heat pumps exchange heat with the ground as opposed to air and water source heat pumps exchange heat with water bodies (Wikipedia contributors, Heat pump., 2023). The choice of heat pump generally depends on the application, location, and cost constraints. Ground source heat pumps are the best choice where cost is not a concern because of their high efficiency and environmental friendliness (Wu, 2009). Air source heat pumps are popular however because of their simple construction and low cost (Wu, 2009) and they rely on air that is abundantly available. This report details the design of a ground source heat pump for space cooling of a single storey detached residential building located in Kampala, Uganda.
Kampala is the capital city of Uganda located in east Africa (0°18'58.61"N, 32°34'55.88"E.). The hottest month in the city is usually January with an average daily high temperature of 28.23°C and a record high temperature of 35°C (Kampala, Uganda Climate., 2023). The daily temperature range for the month is about 9°C. The

cooling system was designed for a desired indoor temperature of 24°C considering an outdoor air temperature of 28°C.

2 METHODOLOGY

2.1 Building description

The system is designed for a three-bedroom single storey house in Kampala, Uganda with a rectangular floor area of 15m by 10m. The floor to ceiling height of the walls is 2.5m. The roof extends to a height of 2m from the level of the ceiling. Below is a schematic of the building.

Figure 1: Schematic of building under consideration.

The windows are assumed to cover 30% of the face area of the building. The windows are also assumed to be double glazed with aluminium frames. It is also assumed that 30% of total window area is facing east, 30% is facing west, 10% is facing north and 10% is facing south. The house is assumed to have a wooden front door and a wooden back door with the total area of both doors being 4m^2. The table below shows the surface area of the walls, the ceiling, the doors, and the windows.

Component	Surface area (m^2)
Windows	$A=\frac{30}{100}\times[2(15\times2.5)+2(10\times2.5)]=37.5$
Ceiling	A=15×10=150
Doors	A=4
Walls	A=(15×10)-(Area of windows + Area of doors) A=83.5

Table 1: Surface area of windows, walls, doors, and ceiling.

The walls are made of 8inch (0.2m) thick fired clay bricks covered with an inch of cement and sand aggregate mixture. The total thickness of the walls in thus 10 inches

(0.25m). The ceiling is assumed to be made of a wooden frame with fibre glass insulation and the attic of the house is vented.
The thermal transmittance values of all the material described above are listed below (ASHRAE, 2021 ASHRAE Handbook—Fundamentals (SI), 2021).

Component	Thermal transmittance (U) W/m^{2}K
Ceiling	0.18
Walls	2.5
Doors	2.61
Windows	3.83

Table 2: U values of building materials.

2.2 Calculations

2.2.1 Cooling load

The cooling load for the building was calculated through the residential load factor method (RLF) described in the 2021 ASHRAE Handbook (ASHRAE, 2021 ASHRAE Handbook—Fundamentals (SI), 2021). The calculations were made considering external heat gain by conduction and radiation through the building envelope, and internal heat gain from occupants and appliances. The heat gains and losses through infiltration were neglected. The heat gains and losses through the floor were also neglected.
The cooling loads due to conduction through the walls, door and ceiling were calculated as follows.

$$Q=A\times CF \qquad 2.1$$

Where:

- Q is the cooling load due to the respective component (walls, ceiling, or doors) in W.
- A is the surface area of the component in m^2.
- CF is the surface cooling factor of the component in W/m^2.

$$CF=U(OF_t\Delta t+OF_b+OF_r DR) \qquad 2.2$$

Where:

- U is the thermal transmittance of the respective component in W/m^{2}K, derived from (ASHRAE, 2021 ASHRAE Handbook—Fundamentals (SI), 2021).
- OF_t, OF_b, and OF_r are opaque surface cooling factors derived from (ASHRAE, 2021 ASHRAE Handbook—Fundamentals (SI), 2021).
- Δt is the design temperature difference in K.
- DR is the daily temperature range in K.

The cooling loads due to conduction and radiation through the windows were also calculated from equation 2.1. However, the surface cooling factor for the windows was calculated as follows.

$$CF=U(\Delta t\text{-}0.46DR)+PXI\times SHGC\times IAC\times FF_s \qquad 2.3$$

Where:

- PXI is the peak exterior irradiance including shading modifications in W/m^2.

$$PXI=T_x\times E_t \qquad 2.4$$

Where:

- T_x is the transmission of the exterior attachment. In this case, there is none, so the value is 1 (derived from (ASHRAE, 2021 ASHRAE Handbook—Fundamentals (SI), 2021)).
- E_t is the peak total irradiance in W/m^2, derived from (ASHRAE, 2021 ASHRAE Handbook—Fundamentals (SI), 2021).

- SHGC is the solar heat gain coefficient, values derived from (ASHRAE, 2021 ASHRAE Handbook—Fundamentals (SI), 2021).
- IAC is the interior shading attenuation coefficient, values derived from (ASHRAE, 2021 ASHRAE Handbook—Fundamentals (SI), 2021).
- FF_s is the fenestration solar load factor, values derived from (ASHRAE, 2021 ASHRAE Handbook—Fundamentals (SI), 2021).

The internal load consisting of the sensible and latent heat gains from the occupants, lights and appliances was estimated as follows.

$$Q_s = 136 + 2.2A_F + 22N \qquad 2.5$$

Where:

- Q_S is the sensible heat gain from occupants, appliances, and lights in W.
- A_F is the total floor area of the building in m^2.
- N is the number of people in the building, assumed to be five for this calculation.

$$Q_l = 20 + 0.22A_F + 12N \qquad 2.6$$

Where:

- Q_l is the latent heat gain from the occupants, appliances, and lights in W.

Equations 2.5 and 2.6 were derived from (ASHRAE, 2021 ASHRAE Handbook—Fundamentals (SI), 2021) and (Hendron, 2005).

The total calculated cooling load is the capacity of the system's evaporator.

2.2.2 Refrigerant circuit

The refrigerant properties were modelled based on an ideal vapour compression refrigeration cycle. Figure 2: Pressure/enthalpy and temperature entropy relationships of refrigerant in ideal vapour compression cycle. shows the pressure/enthalpy and temperature/entropy relationships of the refrigerant at the different points in the cycle. The saturation lines show the saturation state of the refrigerant at each point. At point 1, the refrigerant is a saturated vapour leaving the evaporator, at point 2, it is a super-heated vapour leaving the compressor,

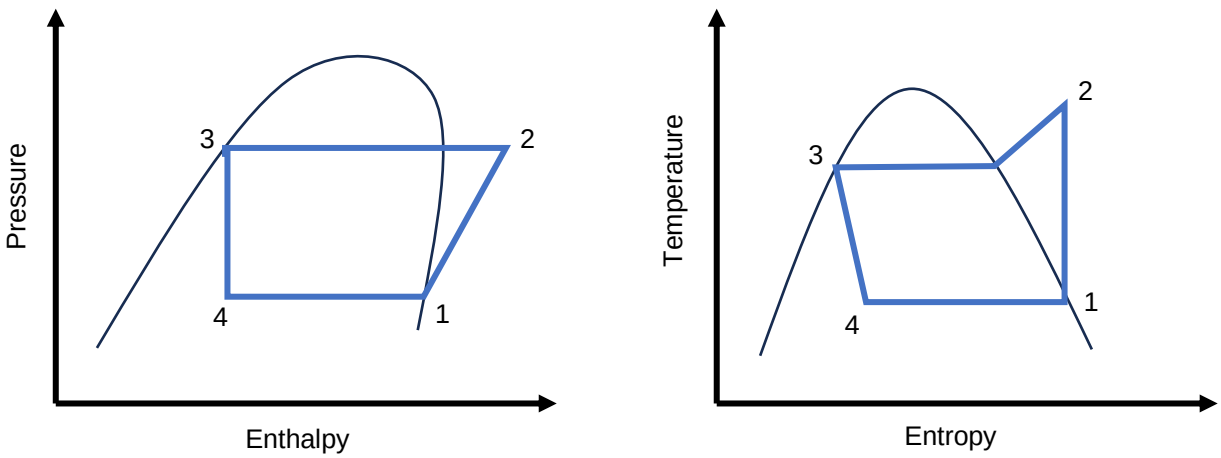

Figure 2: Pressure/enthalpy and temperature entropy relationships of refrigerant in ideal vapour compression cycle.

at point 3, it is a saturated liquid leaving the condenser and at point 4 it is a liquid-vapour mixture leaving the expansion valve.

Refrigerant R134A was chosen for this design. The temperature of the refrigerant in the evaporator must be lower than the temperature of the air flowing over it to allow for heat transfer from the air to the refrigerant. The saturation temperature of the refrigerant in the evaporator was thus selected as 21.6°C from the R134A thermodynamic properties chart (Evans, 2017) based on the design temperature of 24°C and a temperature difference of 4°C across the evaporator.
The compressor was assumed to have a compression ratio of 3 such that the pressure at its outlet is three times the pressure at its inlet.
Based on the relationships outlined in Figure 2: Pressure/enthalpy and temperature entropy relationships of refrigerant in ideal vapour compression cycle., the temperature, enthalpy, and entropy of the refrigerant at the different points in the cycle were determined from the refrigerant properties chart (Evans, 2017). The entropy at point 4 is not necessary for the calculations.
The enthalpy determined at points 1 and 4 (h_1 and h_4) were used to calculate the mass flow rate of the refrigerant from the equation below.

$$Q=m(h_1-h_4) \qquad 2.7$$

Where:

- Q is the calculated cooling load in kW.
- m is the mass flow rate of the refrigerant in kg/s.
- h_1 and h_4 are enthalpy of the refrigerant at points 1 and 4 in the cycle respectively in kJ/kg.

2.2.3 Condenser capacity

The calculated mass flow rate of the refrigerant and its enthalpy at points 2 and 3 (h_2 and h_3) were then used to work out capacity of the condenser using the equation below.

$$Q_R=m(h_3-h_2) \qquad 2.8$$

Where:

- Q_R is the capacity of the condenser in kW.
- h_2 and h_3 are the enthalpy of the refrigerant at points 2 and 3 in kJ/kg.

2.2.4 Compressor

2.2.4.1 Work input

The calculated mass flow rate of the refrigerant and its enthalpy at points 1 and 2 (h_1 and h_2) were used to calculate the work done by the compressor on the refrigerant using the equation below.

$$W_c=m(h_2-h_1) \qquad 2.9$$

Where:

- W_c is the work done by the compressor in kW.
- h_2 and h_1 are the enthalpy values of the refrigerant at points 2 and 1 respectively in kJ/kg.

2.2.4.2 Power consumption

A compressor of 80% efficiency was decided upon for the application so the power consumption of the compressor was calculated using the following equation.

$$\eta = \frac{Work\ Output}{Power\ Consumption} \qquad 2.10$$

$$Power\ Consumption = \frac{Work\ Output}{\eta} \qquad 2.11$$

Where:

- $\boldsymbol{\eta}$ is the efficiency of the compressor.

2.2.5 Coefficient of performance (COP)

The system's COP was calculated from the equation below.

$$\mathrm{COP} = \frac{\text{Cooling load}}{\text{Compressor power consumption}} \qquad 2.12$$

2.3 Ground source heat exchanger (GSHE)

A vertical loop ground source heat exchanger was decided upon for the application as it requires less land area than horizontal loop systems. It also can benefit from more stable temperatures at greater depths in the ground where fluctuations are minimal.

The GSHE consists of 25mm single U-tube high density polyethylene pipes (DR 11 HDPE) bored into the ground. A pump circulates water through the circuit. The boreholes are 125mm in diameter and are filled with thermally enhanced grout. The overall thermal conductivity of the ground and the borehole components is 3.8W/m.K (ASHRAE, 2019). The boreholes are arranged in a 2 by 3 grid layout with 5m spacing between them. There are 6 boreholes in total and the bore field covers an area of 50m^2.

No antifreeze additive is required for the water as the ground temperature at this location is sufficiently high to keep the water in a liquid state.

The boreholes produce water at a temperature of 23°C that picks up heat from the refrigerant circuit and returns to the ground at a temperature of approximately 27°C. Figure 3: Schematic of ground source heat pump. shows a schematic of the system.

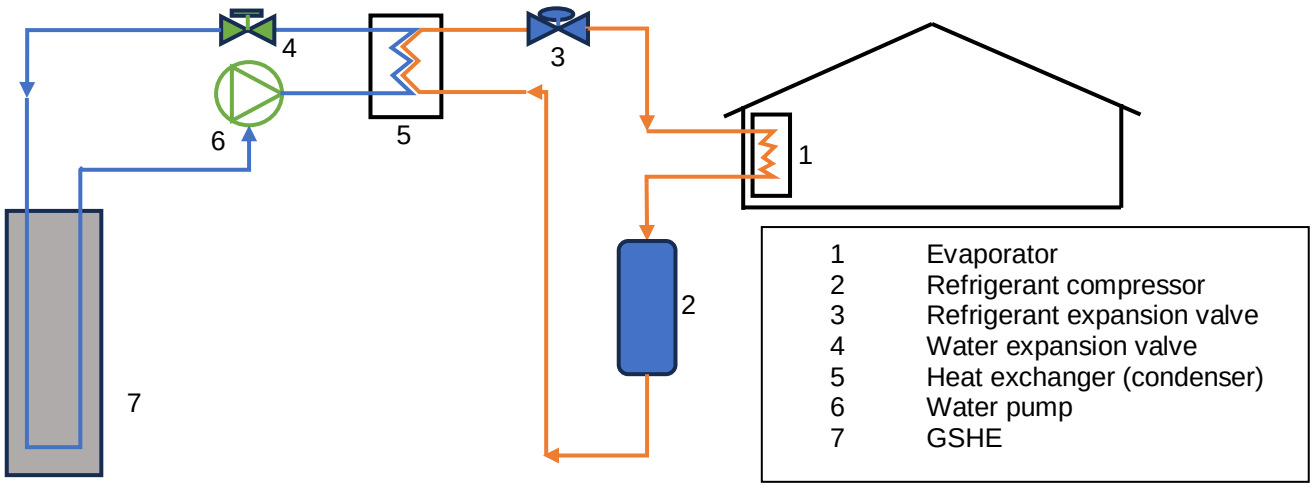

Figure 3: Schematic of ground source heat pump.

The heat rejection rate of the GSHE is the calculated condenser capacity of the refrigerant circuit. The mass flow rate of the water was calculated using the equation below (Primo, 2020).

$$Q = mC\Delta T \qquad 2.13$$

$$m = \frac{Q}{C\Delta T} \qquad 2.14$$

Where:

- Q is the heat transferred to the ground in kW.
- m is the mass flow rate of the water in kg/s.
- C is the specific heat capacity of the water (4.181kJ/kg. K at room temperature)
- ΔT is the change in temperature of the water in K.

The required borehole length was calculated using the concentric cylinder source method as described in the equation below (ASHRAE, 2019).

$$L=\frac{QR_t}{(T_w-T_g)} \qquad 2.15$$

Where:

- L is the required borehole length in m.
- Q is the heat transfer rate (condenser capacity) in W.
- T_W is the temperature of the water returning to the ground in °C.
- T_g is the temperature of the ground in °C.
- R_t total thermal resistance of the ground and borehole components in mK/W (inverse of thermal conductivity).

The depth of each borehole was calculated by dividing the total borehole length by the number of boreholes.

$$\text{Depth of boreholes}=\frac{\text{Borehole length}}{\text{Number of boreholes}} \qquad 2.16$$

The water pump power was selected based on sizing recommendations outlined in 2019 ASHRAE Handbook—HVAC Applications (SI) (ASHRAE, 2019).

2.4 Cost estimate

The installation costs for the system were based on prices from industry service providers in the United Kingdom as there aren't many ground source heat pump service providers in Uganda.

a. The cost of the heat pump was based on prices from (The Heat Pump Warehouse., 2023)
b. The cost of the water pump was based on prices from (Anchor Pumps, 2023).
c. The cost of the boreholes was based on estimates by (IMS Heat Pumps, 2019)

3 RESULTS

3.1 Cooling load

The table below shows the contribution of the ceiling, walls, and doors to the total cooling load.

Component	U (W/m²/K	OF_t	OF_b	OF_r	CF (W/m²)	A (m²)	Q (W)
Ceiling	0.18	0.62	5.51	-0.19	1.1304	150	169.56
Walls	2.5	1	8.2	-0.36	22.4	83.5	1870.4
Doors	2.61	1	8.2	-0.36	23.3856	4	93.5424

Table 3: Cooling loads due to walls, ceiling, and doors.

The table below shows the contribution to the cooling load from the windows facing east, west, north, and south.

Direction	U (W/m^2K)	T_x	E_t (W/m^2)	PXI	SHGC	IAC	FFs	CF (W/m^2K)	A (m^2)	Q (W)
East	3.83	1	734	734	0.37	0.64	0.31	53.34527	11.25	600.1343
West	3.83	1	734	734	0.37	0.64	0.56	96.79807	11.25	1088.978
North	3.83	1	172	172	0.37	0.64	0.44	17.38482	7.5	130.3862
South	3.83	1	391	391	0.37	0.64	0.47	42.98054	7.5	322.354

Table 4: Cooling load due to windows.

The estimated contribution to the cooling load from the occupants, lights and appliances is as follows:

$$Q_{sensible}=576W$$
$$Q_{latent}=113W$$

From the results above, the total cooling load of the house was calculated as 4964.355W or 4.96kW.

3.2 Refrigerant circuit

3.2.1 Refrigerant properties

The tables below show the properties of the refrigerant at each point in the refrigeration cycle.

Point 1 (Saturated vapour)

Temperature (°C)	21.6
Pressure (Kpa)	600
Enthalpy (kJ/kg)	262.4
Entropy (kJ/kg.K)	0.9219

Point 2 (Super-heated vapour)

Temperature (°C)	67
Pressure (Kpa)	1800
Enthalpy (kJ/kg)	284.9
Entropy (kJ/kg.K)	0.9219

Point 3 (Saturated liquid)

Temperature (°C)	62.9
Pressure (Kpa)	1800
Enthalpy (kJ/kg)	144.1
Entropy (kJ/kg.K)	0.5031

Point 4 (Liquid-vapour mixture)

Temperature (°C)	21.6
Pressure (Kpa)	600
Enthalpy (kJ/kg)	144.1

Table 5: Refrigerant properties at the different points of the refrigeration cycle.

The mass flow rate of the refrigerant from equation 2.7 is shown below.

$$Q=m(h_1\text{-}h_4)$$
$$4.96=m(262.4\text{-}144.1)$$
$$m=0.042kg/s$$

3.2.2 Condenser capacity

From equation 2.8:

$$Q_R=m(h_3\text{-}h_2)$$
$$Q_R=0.042(144.1\text{-}284.9)$$
$$Q_R=\text{-}5.914kW$$

The negative value indicates that heat is being rejected by the condenser.

3.2.3 Compressor work input, and power consumption

From equation 2.9:

$$W_c = m(h_2 - h_1)$$
$$W_c = 0.042(284.9 - 262.4)$$
$$W_c = 0.945\text{kW}$$

From equation 2.11:

$$Power\ Consumption = \frac{0.945}{0.8}$$

$$Power\ Consumption = 1.18\text{kW}$$

3.2.4 Coefficient of performance (COP)

From equation 2.12:

$$\text{COP} = \frac{4.96}{1.18}$$
$$\text{COP} = 4.2$$

3.3 Ground source heat exchanger (GSHE)

The mass flow rate of the water flowing through the GSHE from equation 2.14 is shown below.

$$m = \frac{5.914}{4.181 \times 4}$$
$$m = 0.354\text{kg/s}(0.47\text{L/s})$$

A 0.12kW water pump was selected for the application. 2019 ASHRAE Handbook—HVAC Applications (SI) recommends using a 0.12kW pump to deliver flow rates of 0.5 to 0.6 L/s for ground source heat pumps with nominal heat pump capacities up to 10.5kW (ASHRAE, 2019).

Required borehole length as calculated from equation 2.15 is:

$$L = \frac{5914 \times 0.263}{4}$$
$$L = 389.079\text{m}$$

The required borehole length is approximately 390m.

$$\text{Depth of boreholes} = \frac{390}{6}$$
$$\text{Depth of boreholes} = 65\text{m}$$

3.4 Estimated cost

The table below shows a breakdown of the estimated installation costs for the system.

Item	Number of Units	Unit Price	Total
Heat pump	1	£6000	£6000
Bore holes	6 (65m each)	£65 per metre	£25350
Water pump	1	£800	£800
Total			£32150

4 Conclusion

Although the ground source heat pump has a higher installation cost than an air source heat pump for the same application, it is more energy efficient and lasts longer with minimal maintenance, so it is a worthy investment.

The ground temperature is closer to the desired temperature in the house so a ground source heat pump would cool the house at a lower energy cost than a comparable air source heat pump. An air source heat pump in this case would consume more energy because it would be working against a larger temperature gradient between the outside ambient air temperature and the desired temperature in the house.

The energy costs in this case can be offset even further by coupling the ground source heat pump with a renewable energy source like solar photovoltaic technology.

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